

Problem A. Lawn Mopping

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Given N integers $E_1, E_2, \dots, E_N$ and an integer K , select some of them to maximize the total sum, under the constraint that no more than K consecutive elements can be chosen.

Input

The first line contains two integers N ($1 \leq N \leq 10^5$) and K .

The next N lines each contain one integer E_i ($0 \leq E_i \leq 10^9$).

Output

Output a single integer — the maximum possible total sum satisfying the constraint.

Example

standard input	standard output
5 2 1 2 3 4 5	12

Problem B. Longest Increasing Subsequence

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given a sequence of n positive integers, output the length of the longest increasing subsequence in this sequence.

A longest increasing subsequence refers to a subsequence obtained by selecting some numbers from the original sequence in order, such that the selected numbers are **strictly** increasing.

Input

The first line contains n ($1 \leq n \leq 10^5$).

The second line contains n integers, each not exceeding 10^6 , indicating the sequence.

Output

Output the answer.

Example

standard input	standard output
5 3 1 2 4 5	4

Problem C. Treasure

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Teacher JB has found infinite treasure in a cave after years of searching. Now he has been organizing the treasures and trying to select to take away some of them. He finds that each type of treasure has one or more identical items. After roughly estimating the value of each type, he begins the selection process.

Teacher JB has a collecting cart with a maximum load capacity of W . There are n types of treasures in total. For each type i , its value is v_i , its weight is w_i , and there are m_i items available.

Teacher JB wants to choose some treasures to put into his cart so that the total value is maximized, without exceeding the cart's weight limit.

Input

The first line contains two integers n ($1 \leq n \leq 100$) and W ($0 \leq W \leq 4 \times 10^4$), representing the number of treasure types and the maximum load capacity of the cart, respectively.

The next n lines each contain three integers v_i , w_i , and m_i , representing the value, weight, and quantity of the i -th type of treasure.

We also guarantee that $\sum m_i \leq 10^5$. $v_i, w_i \leq 10^9$.

Output

Output a single integer — the maximum total aesthetic value that Teacher JB can achieve within the available time.

Example

standard input	standard output
4 20 3 9 3 5 9 1 9 4 2 8 1 3	47

Note

Output a single integer — the maximum total value of treasures that can be collected without exceeding the cart's capacity.

Problem D. King Placement

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

On an $N \times N$ chess chessboard, place K kings such that none of them attack each other.

A king can attack the eight surrounding squares - one in each of the following directions: up, down, left, right, and the four diagonals (upper-left, upper-right, lower-left, lower-right).

Input

A single line containing two integers $N(1 \leq N \leq 9)$ and $K(1 \leq K \leq N \times N)$.

Output

Output the total number of valid arrangements.

Example

standard input	standard output
3 2	16

Problem E. Merging Stones

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Place n piles of stones around a circular playground. The goal is to merge all the stones into a single pile in an orderly manner. Each time, you may only choose two adjacent piles to merge into a new pile, and the number of stones in the new pile is recorded as the score of that merge.

Write a program that reads the number of piles n and the number of stones in each pile, and computes the following:

The maximum possible total score after performing $n - 1$ merges.

The minimum possible total score after performing $n - 1$ merges.

Input

The first line contains an integer n ($1 \leq n \leq 200$), the number of stone piles. The second line contains n integers, representing the number of stones in each pile.

Output

The first line: the minimum total merge score.

The second line: the maximum total merge score.

Example

<code>standard input</code>	<code>standard output</code>
4	43
4 5 9 4	54